

CSIR-UGC NATIONAL ELIGIBILITY TEST

MATHEMATICAL SCIENCES

Preparation Blueprint

Based on Complete Analysis of
December 2025 & June 2025 Papers
All 240 questions classified, mapped & prioritised



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1 Exam Structure at a Glance

Exam Format

Duration: 3 hours (180 minutes) **Total Marks:** 200 **Mode:** Online (CBT)
All three parts must be attempted in one sitting. Negative marking applies in all parts.

Part	Type	Total Qs	Attempt	+Marks	−Marks	Max Score
Part A	MCQ	20	Any 15	+2	−0.5	30
Part B	MCQ	40	Any 25	+3	−0.75 / −1	75
Part C	MSQ	60	Any 20	+4.75	−1.25	95
Total		120	60			200

MSQ Warning

Part C is Multiple Select Questions (MSQ). **Any single wrong option selected** incurs −1.25 penalty *and* forfeits the full +4.75 credit. Never guess in Part C.

1.1 Score Targets

Goal	Part A (of 30)	Part B (of 75)	Part C (of 95)	Total
JRF	26–28	60–66	85–90	~175–184
Lectureship	22–26	51–60	70–80	~150–170
Safe Pass	20–22	45–51	60–70	~130–145

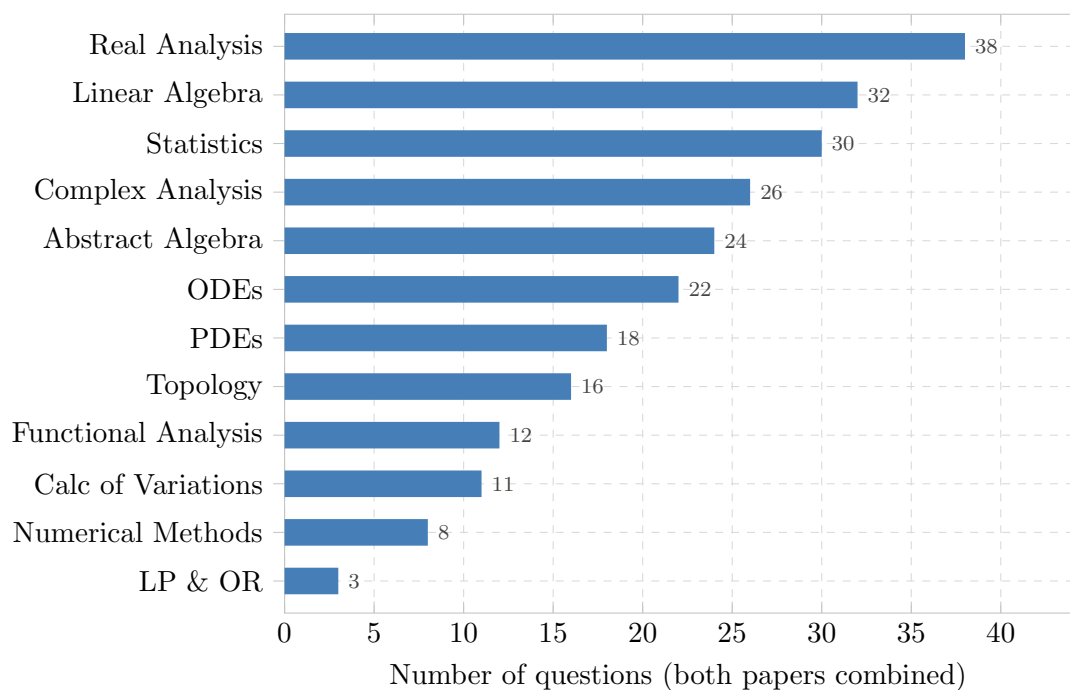
Cutoffs vary by year. JRF cutoff for Mathematical Sciences is typically among the highest across subjects.

2 Topic-wise Question Distribution

2.1 Parts B & C Combined (200 questions across both papers)

The following analysis covers all 200 mathematics questions (Part B: 80 total, Part C: 120 total) across both the December 2025 and June 2025 papers.

Topic	Dec 2025	Jun 2025	Total	Priority
Real Analysis	19	19	38	Tier 1
Linear Algebra	16	16	32	Tier 1
Statistics & Probability	15	15	30	Tier 1
Complex Analysis	13	13	26	Tier 2
Abstract Algebra	12	12	24	Tier 2
Ordinary Diff. Equations	11	11	22	Tier 2
Partial Diff. Equations	9	9	18	Tier 3
Topology	8	8	16	Tier 3
Functional Analysis	6	6	12	Tier 3
Calculus of Variations	5–6	5–6	11	Tier 4
Numerical Methods	4	4	8	Tier 4
Linear Programming	1–2	1–2	3	Tier 4
Total	~100	~100	200	



2.2 Part A — General Aptitude (20 questions per paper)

Sub-topic	Dec 2025	Jun 2025
Mathematical reasoning, arithmetic & ratio problems	4	4
Number theory, sequences, combinatorics	3	3
Logical reasoning (statements, blood relations, coding)	3	3
Data interpretation (pie charts, bar graphs)	3	3
Geometry, distance, area, volume	2	2
Probability & basic statistics	2	2
Science-based / environmental reasoning	2	2
Current affairs framed as reasoning	1	1
Total	20	20

3 Priority Topic Deep-Dives

Key Insight

Master Tiers 1 and 2 (six topics) and you cover approximately **75%** of all Part B and Part C marks. The sub-topics below are exactly what appeared across both 2025 papers.

3.1 Tier 1 — Master First (Highest ROI)

38 Qs across both papers $\approx 19\%$ of B+C

Why it dominates: Single most tested subject. Questions are deeply conceptual — never purely computational. Appears in every part of every paper.

Sub-topics to master:

- Sequences and series: convergence tests (ratio, root, comparison), Cauchy sequences, $\limsup/\liminf$, rearrangements
- Uniform continuity and the composition of uniformly continuous functions
- Pointwise vs. uniform convergence of sequences of functions; Weierstrass M-test
- Riemann integration: Riemann integrable functions, improper integrals on $[0, \infty)$
- Monotone functions: one-sided limits always exist; surjective $\Rightarrow$ continuous
- Metric space topology: compactness, completeness, distance function $d(x, A)$ (Lipschitz)
- Countability arguments: decimal expansions, Cantor-type sets, image under continuous injections
- Bounded below / above sets; $\inf A$, $\sup A$ computations for tricky sets

Exact questions from both papers (appear in Part C):

Q66 (decimal expansion convergence), Q67 (Riemann integrable on $[0, \infty)$), Q86 (uniform convergence on sub-intervals), Q87 (metric space $d(x, A)$ properties), Q88 (bounded-below set), Q89 (integral function), Q90 (CDF / monotone function), Q92 (fn sequences in $C[0, 1]$), Q96 (monotone fn one-sided limits) — *identical in Dec & Jun 2025*.

32 Qs across both papers $\approx 16\%$ of B+C

Why it matters: Every paper has 4 Part B and 6 Part C Linear Algebra questions. Eigenvalue theory and structural results (projections, quadratic forms) recur without fail.

Sub-topics to master:

- Eigenvalues, trace, determinant, characteristic polynomial; Newton's power-sum identities
- Cayley-Hamilton theorem; minimal polynomial
- Diagonalisability over $\mathbb{R}$ and $\mathbb{C}$; Jordan normal form (when does $T^2 \text{ diag.} \Rightarrow T \text{ diag.}$?)
- Inner product spaces: orthonormality, Gram-Schmidt, orthogonal complements

- Projections: $T^2 = T$ (idempotent), $A^2 = I$ operators and their commutant/anticommutant
- Quadratic forms: positive definite, signature, Sylvester's law
- Rank-nullity; dimension of quotient spaces ($X/\ker \phi$)
- Matrices over finite fields $\mathbb{F}_q$: counting matrices of given rank
- Invariant subspaces in complex vector spaces; existence of eigenvalues over $\mathbb{C}$

Recurring exact questions:

Q74 ($A^2 = I$, $\text{Tr}(A) = 0$), Q76 ($M_n(\mathbb{C})$ with rank- k inner product), Q78 (spanning set, dimension 4 or 5), Q99 (integer matrix eigenvalues, minimal poly), Q103 (T^2 diagonalisable $\Rightarrow T?$), Q107 (4-dim complex VS), Q109 ($T^2 = T$ projection), Q113 (sufficient & ancillary statistics) — *identical in both papers.*

30 Qs across both papers $\approx 15\%$ of B+C

Why it matters: Highest variety of sub-topics. Also most scoring if you specialise — each sub-topic is self-contained and testable independently.

Sub-topics to master:

- *Estimation:* MLE (discrete/continuous), UMVUE, sufficient statistics (Factorisation), ancillary statistics, Cramér-Rao bound
- *Hypothesis testing:* Neyman-Pearson (UMP tests), LRT, sign test, p -values, confidence intervals
- *Bayesian:* Beta-Binomial conjugate, posterior mean/variance with improper priors
- *Distributions:* Exponential, Gamma, Uniform (MOM/MLE), Binomial, Normal (folded, conditional), Poisson, MGF-based identification
- *Multivariate:* $N_p(\mu, \Sigma)$, Wishart distribution, $E(S^{-1})$, partial correlation, multiple correlation
- *Markov chains:* Recurrence, transience, positive recurrence, stationary distribution, mean recurrence time
- *Regression:* Multiple linear regression, R^2 , Adjusted R^2 , F -test, BLUE (Gauss-Markov), BLUEs uncorrelated
- *Sampling:* SRSWR vs. SRSWOR, stratified sampling, unbiased estimators
- *Queuing:* $M/M/c$ systems, steady-state probabilities p_0, p_n
- *Non-parametric:* Sign test, order statistics confidence intervals, survival analysis / hazard rates
- *Probability:* CLT, WLLN, convergence in distribution, sequences of i.i.d. r.v.'s

3.2 Tier 2 — High Priority (Do Not Skip)

26 Qs ≈13%

Sub-topics:

- Holomorphic functions, Cauchy-Riemann equations, analytic continuation
- Cauchy's integral theorem and formula; residue theorem (simple/higher-order poles)
- Contour integrals on specific paths $\gamma_R(t) = Re^{2\pi it}$; computation of $\int_{\gamma} \tan z \, dz$ etc.
- Singularities: poles (order), essential singularities, removable singularities; Laurent series
- Entire functions: Liouville, Picard; growth conditions; image constraints (e.g. $f(\mathbb{C}) \subset \{y = x + 1\}$)
- Möbius (fractional linear) transformations; image of intervals/circles
- Schwarz-Pick lemma: $f : \mathbb{D} \rightarrow \mathbb{D}$ holomorphic, $|f'(a)| \leq \frac{1-|f(a)|^2}{1-|a|^2}$
- Function $g(z) = f(z) - \frac{1}{z}$: pole at 0, no zeros on unit circle, $\max |g| \geq 1$
- Series of holomorphic functions $f_n(z) = \sum_{k=1}^n h(z^k)$; convergence on $\{|z| > 1\}$

24 Qs ≈12%

Sub-topics:

- Group theory: subgroup counting $S(G)$, Sylow theorems, groups of order 8 / 16 / p^n
- Permutation groups S_n ; Burnside's lemma; conjugacy classes
- Galois theory: irreducible polynomial of prime degree p with exactly 2 non-real roots $\Rightarrow G = S_p$
- Irreducibility tests: Eisenstein, reduction mod p ; factorisation in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$
- Ring theory: maximal ideals, prime ideals, PIDs, UFDs, Euclidean domains; rings with divisibility property
- Number theory: Wilson's theorem, Fermat's little theorem, $p \mid (1234)^{p-1} - 1$
- Homeomorphisms of finite topological spaces (tied to group counting)
- Non-abelian groups: $f(a, b) = ab$ not a homomorphism; normal subgroup count

22 Qs ≈11%

Sub-topics:

- IVP existence and uniqueness (Picard-Lindelöf); solutions that are not unique (e.g. $y' = f(y)$ with f not Lipschitz)
- Taylor series solutions about regular and regular singular points (Frobenius method)
- Wronskian: $W(y_1, y_2)$ constant if $p = 0$ in $y'' + p(x)y' + q(x)y = 0$
- Sturm-Liouville problems: eigenvalue sequences λ_n , $\sum 1/\lambda_n$ via Parseval / series identities

- Reduction of order: one solution known, find general solution
- Integral equations: Volterra $y(x) = f(x) + \int_0^x K(x,t)y(t)dt$; Fredholm $y = f + \lambda \int_{-1}^1 K(x,t)y(t)dt$; resolvent kernel
- Oscillation theory: Sturm comparison; zeros of solutions; non-trivial solutions with infinitely many zeros in bounded interval (only if $r(x)$ oscillates)

3.3 Tier 3 — Selective Mastery (Pick Your Best)

Topic	Key sub-topics to cover	Qs
PDEs	d'Alembert (wave), heat equation with Neumann/Dirichlet BCs, Laplace maximum principle, BVP on disk ($u = e^{x+y}$ on $\partial\Omega$), method of characteristics for first-order PDEs, energy methods for $u_{tt} - u_{xx} + u_t = 0$	18
Topology	Product topology (Tychonoff = compact), Hausdorff separation, normality, finite-complement topology (not regular), homeomorphisms of finite spaces, Euclidean topology normality	16
Functional Analysis	Hilbert spaces ℓ^2 ; set $H = \{(a_n) : a_n \leq 1/n\}$ — bounded, convex, not a subspace, no ONB; compact sets in $C[0, 1]$ (Arzelà-Ascoli); operators on Hilbert space	12
Calc of Variations	Euler-Lagrange equation; isoperimetric problems ($\int y^2 dx = 1/7$ constraint); transversality conditions; canonical transformations (generating functions, Hamiltonian mechanics)	11
Numerical Methods	Newton-Raphson: order of convergence p and asymptotic error constant; Gauss-Seidel convergence (spectral radius of iteration matrix); Lagrange interpolation polynomial; cubic splines continuity conditions	8
LP & OR	Simplex method, optimal feasible solution, $M/M/c$ queuing steady-state probabilities, Linear programming in 2 variables with graphical method	3–5

4 Repeat Patterns — Questions Identical Across Both Papers

Critical Finding

18 Part C questions were word-for-word identical in both December 2025 and June 2025 papers. These represent guaranteed high-value preparation targets. Each carries +4.75 marks. Mastering all 18 alone is worth $18 \times 4.75 = \mathbf{85.5}$ marks.

Q#	Topic	Exact structure / key idea tested
Q66	Decimal expansions (RA)	Strictly increasing $(\beta_n) \nearrow \alpha$ (a finite decimal); $b_{n,2025} = a_{2025} - 1$ eventually; digit-by-digit locking
Q67	Riemann integrable functions (RA)	Four statements about $f : [0, \infty) \rightarrow [0, \infty)$; uniformly continuous + convergent integral $\Rightarrow f \rightarrow 0$; existence of counterexample with $f \not\rightarrow 0$
Q68	ODE $y'' + r(x)y = 0$ (ODE)	$r(x) = x^3 \sin(1/x)$; Wronskian = constant; solutions have seq. of zeros $\rightarrow \infty$; no infinitely-many zeros in bounded interval
Q69	Product topology (Topology)	$X = \prod_{c>0} [0, c]$; compact (Tychonoff), Hausdorff, connected; not metrizable
Q74	$A^2 = I$, $\text{Tr}(A) = 0$ (LA)	Anticommutant $\{B : AB + BA = 0\}$ is 8-dim; commutant 8-dim; existence of B with $B^2 = I$, $\text{Tr}(B) = 0$
Q75	$M/M/2$ queue (Statistics)	$\lambda = 4$, $\mu = 1$, capacity 3; $p_0 + p_3 = 17/29$; $p_3 - p_0 = 15/29$
Q76	$M_n(\mathbb{C})$ inner product (LA/FA)	$W_k = \{A : \langle A, B \rangle = 0 \ \forall \text{ rank-}k \ B\}$; $W_1 = W_2 = W_n = \{0\}$; $W_0 \neq \{0\}$
Q77	Borrowing / troublesome (Combinatorics)	6 persons, 12 months, pigeonhole; at least one troublesome in Jan; in any two consecutive months at least one troublesome in one of them
Q78	Spanning sets (LA)	$\{u, v, w, x, y\}$ spans, $\{u, v, x, y\}$ LI; $\dim V \in \{4, 5\}$; $2u - 3v + 5w = 0 \Rightarrow \{v, w, x, y\}$ basis
Q79	Isoperimetric CoV (CoV)	$J(y) = \int_0^1 x^2 (y')^2 dx$, $y(0) = 0$, $y(1) = 1$, $\int y^2 = 1/7$; $y'(1/2) = 3/4$, $y'(1/\sqrt{3}) = 1$
Q80	Multivariate Normal / Wishart (Statistics)	$n = 20$ from $N_{12}(\mu, \Sigma)$; $E(S^{-1}) = \frac{19}{6}\Sigma^{-1}$; $\text{tr}(19\Sigma^{-1}S) \sim \chi_{228}^2$
Q92	$f_n = n(1 - nx)$ in $C[0, 1]$ (FA/RA)	Uniformly bounded; converges pointwise to 0; converges uniformly to 0; set $\{f_n\}$ is compact in $C[0, 1]$

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Q#	Topic	Exact structure / key idea tested
Q93	$g(z) = f(z) - 1/z$ (CA)	f entire; g has pole at 0; no zeros with $ \alpha = 1$; finitely many zeros; $\max_{ z =1} g \geq 1$
Q94	ℓ^2 set $H = \{(a_n) : a_n \leq 1/n\}$ (FA)	Bounded; convex; not a subspace; contains no ONB of ℓ^2
Q108	Galois group $= S_p$ (Algebra)	Irreducible $f \in \mathbb{Q}[x]$ of prime deg. p , exactly 2 non-real roots; $(12) \in G$; p -cycle $\in G$; $G = S_p$
Q109	$T^2 = T$ projection (LA)	$W = \text{span}\{I, T^n\}$; $\dim W = 2$; idempotents in W ; $U^2 = U$, $(T - U)^2 = T - U \Rightarrow (TU)^2 = TU$
Q110	Non-abelian group (Algebra)	$f(a, b) = ab$ not a homomorphism; every divisor has subgroup $\Rightarrow \geq 3$ normal subgroups
Q120	$f(z) = \sin^2 z + \cos^2 z $ (CA)	Not real-valued; not entire; not identically 1; finitely many zeros on imaginary axis

4.1 Recurring Sub-topic Patterns (Across Multiple Questions)

Pattern	What to study
Trace-eigenvalue identities	Newton's identities; $\text{tr}(A) = \sum \lambda_i$, $\text{tr}(A^2) = \sum \lambda_i^2$; char poly from trace data
Uniform vs. point-wise	Weierstrass M-test; sequences f_n on $[0, 1]$; Dini's theorem
Contour integrals on γ_R	Residue theorem at $z = \pi/2$ ($\tan z$), $z = 0$ ($z \sin z$); winding numbers
Beta-Binomial Bayesian	Posterior mean $= \frac{X+\alpha}{n+\alpha+\beta}$; posterior $\geq$ prior mean iff $X/n \geq \alpha/(\alpha+\beta)$
Markov chain classification	States: recurrent / transient / ergodic; stationary $\pi P = \pi$; mean return time
Fredholm integral equations	Degenerate kernels; eigenvalues λ of K ; resolvent $R(x, t; \lambda)$
Möbius image of real line	$f(z) = (z - i)/(z + i)$; images of intervals and lines as arcs / semicircles
Gauss-Seidel convergence	Spectral radius of $T_{GS} = -(D + L)^{-1}U$ must be < 1
Newton-Raphson order	$p = 2$ for simple zero; $p = 1$ for multiple / inflection ($\sin x = 1$ at $\pi/2$)
Cubic splines	C^2 continuity conditions; $2a + b + 2c =$ specific value from both papers

5 Part-wise Question Split Tables

5.1 Part B — MCQ (40 questions per paper, attempt 25)

Topic	Dec 2025	Jun 2025	Expected Range	Recommend attempt?
Statistics & Probability	5	5	5	Yes — all 5
Real Analysis	4	5	4–5	Yes — 4–5
Linear Algebra	4	4	4	Yes — all 4
Complex Analysis	4	4	4	Yes — 3–4
Abstract Algebra	4	4	4	Yes — 3–4
ODEs	3	3	3	Yes — 2–3
PDEs	3	2	2–3	Yes — 2
Topology	2	2	2	Selective
Numerical Methods	2	2	2	Yes — if prepared
Calculus of Variations	2	2	2	Yes — if prepared
Functional Analysis	1	1	1	If fast to solve
Linear Programming	1	1	1	Yes — easy marks
Total Part B	35–40	35–40	40	<i>Attempt: 25</i>

5.2 Part C — MSQ (60 questions per paper, attempt 20)

Topic	Dec 2025	Jun 2025	Expected Range	Recommend attempt?
Real Analysis	9	8	8–9	Select 4–5
Statistics & Probability	8	8	8	Select 4–5
Linear Algebra	6	6	6	Select 3–4
Complex Analysis	6	5	5–6	Select 2–3
Abstract Algebra	5	5	5	Select 2–3
ODEs	4	4	4	Select 1–2
PDEs	4	4	4	Select 1–2
Topology	4	4	4	Select 1–2
Functional Analysis	3	3	3	Select 1
Calculus of Variations	3	3	3	Select 1
Numerical Methods	2	2	2	Selective
Classical Mechanics	2	2	2	Selective
Combinatorics / Logic	2	1	1–2	Selective
Total Part C	58–60	55–60	60	<i>Attempt: 20</i>

6 12-Week Preparation Schedule

Plan Overview

Four phases: **Weeks 1–4** build the top-3 topics deep. **Weeks 5–8** cover the next three topics. **Weeks 9–10** tackle remaining topics. **Weeks 11–12** are full timed mock tests with gap analysis. Adjust based on existing knowledge.

Phase 1 — Foundation (Weeks 1–4): Real Analysis + Linear Algebra

Real Analysis I — Sequences, Series, Continuity

Convergence tests (ratio, root, comparison), Cauchy sequences, $\limsup / \liminf$, **W1** Dirichlet summability, uniform continuity definition and properties, Weierstrass M-test. *Solve:* Dec Q33, Q36, Q39, Q47, Q49; Jun Q24, Q25, Q26, Q56. **Target:** 10+ hours.

Real Analysis II — Metric Spaces, Integration, Monotone Functions

Metric space compactness, $d(x, A)$ Lipschitz continuity, Riemann integrable functions on $[0, \infty)$, monotone function one-sided limits, $\sup/\inf$ of tricky sets, countability. *Solve all 9 identical Part C questions:* Q66–Q68, Q86–Q90, Q92, Q96. **Target:** 10+ hours.

Linear Algebra I — Eigenvalues, Trace, Diagonalisability

W3 Newton's power-sum identities, char poly from trace data, Cayley-Hamilton, diagonalisability criteria, minimal polynomial, Jordan normal form basics. *Solve:* Dec Q23, Q25, Q45, Q53; Jun Q27, Q28, Q29, Q30. **Target:** 10 hours.

Linear Algebra II — Projections, Inner Products, Quadratic Forms, Finite Fields

W4 T and $A^2 = I$ structures, $M_n(\mathbb{C})$ inner product, W_k spaces, vector space spanning dimension argument, quadratic form signature, matrices over $\mathbb{F}_q$. *Solve identical C questions:* Q74, Q76, Q78, Q99, Q103, Q107, Q109. **Target:** 10 hours.

Phase 2 — Core Expansion (Weeks 5–8): Algebra + Complex Analysis + Statistics

Complex Analysis — Holomorphic Functions, Singularities, Contour Integrals

W5 Cauchy's theorem, residue theorem, classification of singularities, Taylor/Laurent series, entire functions (Liouville, Picard), contour integrals on $\gamma_R(t) = Re^{2\pi it}$. *Solve:* Dec Q28, Q41, Q42, Q44; identical C Qs Q93, Q120. Also Schwarz-Pick (Q101). **Target:** 10 hours.

Abstract Algebra — Groups, Rings, Galois Theory

W6 Sylow theorems, groups of order 8, Galois group = S_p (irreducible poly, 2 non-real roots), Eisenstein criterion, maximal/prime ideals, rings with divisibility, Wilson/Fermat. *Solve:*

W6 Q22, Q30, Q60, Q62, Q65; identical C Qs Q108, Q110. **Target:** 10 hours.

Statistics I — Estimation, Testing, Bayesian

W7 MLE, sufficient statistics (Factorisation theorem), ancillary statistics, UMP tests (Neyman-Pearson), LRT, Beta-Binomial conjugate, confidence intervals for various distributions. *Solve:* Dec Q21, Q51, Q54, Q55, Q57; Jun Q53, Q54, Q55, Q56. **Target:** 12 hours.

Statistics II — Multivariate, Markov Chains, Regression, Sampling

W8 Wishart distribution, $E(S^{-1})$, partial correlation, Markov chain classification (recurrence/-transience), multiple regression R^2/F -test, SRSWOR/stratified sampling, $M/M/c$ queuing, sign test. *Solve identical C Qs:* Q75, Q77, Q80, Q97, Q98. **Target:** 12 hours.

Phase 3 — Secondary Topics (Weeks 9–10)

ODEs + PDEs + Integral Equations

W9 Frobenius method, Sturm-Liouville $\sum 1/\lambda_n$, Wronskian constancy, Fredholm/Volterra integral equations, resolvent kernel; PDE: d'Alembert, heat equation with BCs, Laplace (maximum principle), method of characteristics, energy method. *Solve identical Qs:* Q68, Q72, Q79, Q81, Q84, Q116, Q118. **Target:** 12 hours.

Topology + Functional Analysis + CoV + Numerical Methods

W10 Product topology Tychonoff, normality, ℓ^2 Hilbert space (set H), compact sets in $\mathbb{R}^n$, Euler-Lagrange with isoperimetric constraints, Newton-Raphson order-2 convergence, Gauss-Seidel spectral radius, Lagrange interpolation, cubic splines. *Solve identical Qs:* Q69, Q71, Q79, Q82, Q92, Q94. **Target:** 10 hours.

Phase 4 — Mock Tests & Final Revision (Weeks 11–12)

Full Mock — December 2025 Paper (timed: 180 minutes)

W11 Part A: 20 min. Part B: 60 min. Part C: 100 min. Record which questions you got wrong and why. Categorise errors: conceptual gap / calculation error / misread / time pressure. Identify weakest 2–3 sub-topics. **Target:** 1 full test + 4 hrs gap analysis.

Full Mock — June 2025 Paper + Targeted Revision

W12 Attempt June 2025 under strict timed conditions. Revise weakest sub-topics from Week 11 analysis. Final day: revise all 18 identical Part C questions (maximum ROI per hour). Prepare a one-page *cheat sheet* of key theorems for last-day review. **Target:** 1 full test + 4 hrs targeted revision.

7 Exam Day Strategy

7.1 Part A — General Aptitude (20 minutes recommended)

- **Attempt only 15 of 20.** Score if all 15 correct: $15 \times 2 = 30$. If you attempt 15 correct + 5 wrong: $30 - 2.5 = 27.5$. Never guess.
- **Recommended order:** Arithmetic/ratio problems first (fastest) $\rightarrow$ Logical reasoning $\rightarrow$ Data interpretation $\rightarrow$ Geometry/science (slowest).
- **Time rule:** ≤ 75 seconds per question. If stuck after 45 seconds, skip and return.
- Both 2025 papers had 8 arithmetic + 6 logic questions — these are the most predictable category.

7.2 Part B — MCQ (60 minutes recommended)

- **Scan all 40 questions first (5 minutes).** Mark each: $\checkmark$ (very confident), ? (partial), $\times$ (skip).
- Solve all $\checkmark$ first, then revisit ? with remaining time. Never attempt $\times$ unless surplus time and near-certainty.
- **Attempt exactly 25 — never more.** Penalty is -0.75 or -1 per wrong answer.
- Recommended topic-wise count: Statistics (5) + Real Analysis (4–5) + Linear Algebra (4) + Complex Analysis (3–4) + Abstract Algebra (3–4) + ODEs/PDEs (3) + LP/Numerical (2) = 24–26 total. Choose your strongest.
- ≈ 2.4 minutes per question. Statistics and LP tend to be fastest; Topology and Functional Analysis slowest.

7.3 Part C — MSQ (100 minutes recommended)

The 90% Confidence Rule

In Part C, only select an option if you are **at least 90% confident** it is correct. If you know options 1 and 3 are correct but are unsure about option 2 — select only 1 and 3. The -1.25 penalty for any wrong option destroys both the penalty cost and the full $+4.75$ credit simultaneously.

- **Attempt exactly 20 questions.** Target: 18–19 fully correct = 85.5–90.25 marks from Part C alone.
- **Prioritise the 18 identical questions** (Q66–Q120 range). You already know their exact structure and correct answers from preparation. These are free marks.
- **Recommended selection order:** Real Analysis (pick 4) $\rightarrow$ Statistics (pick 4) $\rightarrow$ Linear Algebra (pick 3) $\rightarrow$ Complex Analysis (pick 3) $\rightarrow$ Abstract Algebra (pick 2) $\rightarrow$ ODEs/PDEs/-Topology (pick remaining 4).
- For each question: identify the options you are *certain* are correct and the options you are *certain* are wrong. Select only the former. Leave unresolved options unselected.
- **Time management:** Read every option carefully. An average of 5 minutes per question is appropriate — some will take 2 minutes, some 8 minutes.

7.4 Common Traps to Avoid

Trap	How to avoid it
Selecting a “seems right” MSQ option	Apply the 90% rule. If you have any doubt, leave it unselected.
Attempting 26+ questions in Part B	The 26th attempt on a wrong answer costs you 1 mark net. Stop at 25.
Confusing pointwise vs. uniform convergence	Uniform: $\sup_x f_n(x) - f(x) \rightarrow 0$. Pointwise: only fix x . Completely different!
Missing “NECESSARILY TRUE” qualifier	A statement must hold for <i>all</i> instances in the given class, not just one example.
FALSE vs. NOT FALSE questions	Jun Q32, Q33, Q38 ask which statement is FALSE . Invert your reasoning.
Misreading Möbius image	Always compute what happens to 3 points on the domain. Use the circle/line preservation property.
Part C: selecting all 4 options reflexively	When you’re 90% sure about 3 options and 50/50 on the 4th — skip the 4th.
Skipping Part C entirely for a topic	Even one fully-correct answer in a weak topic (4.75 marks) beats 2 wrong Part B attempts (−2 marks).

8 Recommended Study Resources

Topic	Primary Reference	Supplementary
Real Analysis	W. Rudin, <i>Principles of Mathematical Analysis</i>	Apostol, <i>Mathematical Analysis</i>
Linear Algebra	S. Axler, <i>Linear Algebra Done Right</i>	Hoffman & Kunze, <i>Linear Algebra</i>
Abstract Algebra	Dummit & Foote, <i>Abstract Algebra</i>	Herstein, <i>Topics in Algebra</i>
Complex Analysis	L. Ahlfors, <i>Complex Analysis</i>	Conway, <i>Functions of One Complex Variable</i>
Topology	J.R. Munkres, <i>Topology</i>	Simmons, <i>Introduction to Topology</i>
ODEs	Coddington & Levinson, <i>ODE Theory</i>	Birkhoff & Rota, <i>Ordinary Differential Equations</i>
PDEs	L.C. Evans, <i>Partial Differential Equations</i>	Strauss, <i>PDEs: An Introduction</i>
Functional Analysis	E. Kreyszig, <i>Introductory FA</i>	Simmons, <i>Introduction to FA</i>
Statistics	Hogg, McKean & Craig, <i>Mathematical Statistics</i>	Casella & Berger, <i>Statistical Inference</i>
Probability	S.M. Ross, <i>Introduction to Probability Models</i>	Billingsley, <i>Probability & Measure</i>
Calculus of Variations	I.M. Gelfand & Fomin	Elsgolts, <i>Differential Equations & CoV</i>
Numerical Methods	S.S. Sastry, <i>Introductory Methods of Numerical Analysis</i>	Burden & Faires, <i>Numerical Analysis</i>

8.1 Practice Material

- **Past papers (highest priority):** Solve last 10 years of CSIR papers in strict timed conditions. The Dec 2025 and Jun 2025 papers in this document are the most recent and most predictive.
- **Official source:** NTA website (csirnet.nta.nic.in) for official answer keys and previous papers.
- **Topic-specific banks:** NBHM PhD question papers for Real Analysis, Linear Algebra, and Complex Analysis contain questions of comparable depth.
- **CMI/ISI entrance exams:** Excellent supplementary practice for Abstract Algebra and Real Analysis.

9 Quick-Reference Summary Sheet

At-a-Glance: What to Study, In What Order

1. **Real Analysis** (Week 1–2): *uniform convergence, metric spaces, Riemann integration, monotone functions, countability*
2. **Linear Algebra** (Week 3–4): *char poly, Cayley-Hamilton, $T^2 = T/A^2 = I$, inner products, W_k spaces, $\mathbb{F}_q$ matrices*
3. **Complex Analysis** (Week 5): *residue theorem, $g(z) = f(z) - 1/z$, Möbius maps, Schwarz-Pick, entire functions*
4. **Abstract Algebra** (Week 6): *Galois $G = S_p$, groups of order 8, irreducibility (Eisenstein), maximal ideals*
5. **Statistics** (Week 7–8): *MLE, UMP tests, Bayesian (Beta-Binomial), Wishart, Markov chains, $M/M/2$ queue, multiple regression*
6. **ODEs + PDEs** (Week 9): *Sturm-Liouville, Frobenius, integral equations, d'Alembert wave, heat BVP, energy method*
7. **Topology + FA + CoV + Numerical** (Week 10): *product topology, ℓ^2 , Euler-Lagrange, N -R order 2, Gauss-Seidel*
8. **Mock tests** (Week 11–12): *Dec 2025 paper $\rightarrow$ gap analysis $\rightarrow$ Jun 2025 paper $\rightarrow$ final revision of 18 identical Qs*

Part C non-negotiables (memorise these): $M/M/2$ queue ($\lambda = 4$, $\mu = 1$, cap 3) · Product topology $\prod [0, c]$ (compact, Hausdorff, connected, not metrizable) · $A^2 = I$, $\text{Tr}(A) = 0$: anticommutant is 8-dim · Galois: irred. prime deg., 2 non-real roots $\Rightarrow G = S_p$ · $g(z) = f(z) - 1/z$: pole at 0, no zero on $|z| = 1$ · $f_n = n(1 - nx)$: compact in $C[0, 1]$, converges uniformly to 0 · $H = \{(a_n) : |a_n| \leq 1/n\} \subset \ell^2$: bounded + convex, not a subspace · $f(z) = \sin^2 z + \cos^2 |z|$: not entire, finitely many zeros on imaginary axis

Final Motivation

The 18 identical questions alone are worth **85.5 marks out of 200**. The top 6 topics cover **~75% of Part B+C marks**. A focused, topic-prioritised approach with strong Part C discipline is the single most effective strategy for CSIR NET Mathematical Sciences.

Good luck!

Blueprint based on CSIR-UGC NET Mathematical Sciences — December 2025 (Code 704, Shift 2) & June 2025 (Code 704, Shift 2)

NTA Final Answer Keys used for all 240 questions. Compile with pdf_lat_ex (run twice for table of contents).